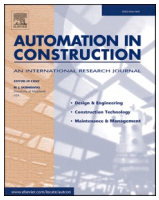
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# Autonomous dimensional inspection and issue tracking of rebar using semantically enriched 3D models

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ABSTRACT

Accurate and efficient inspection of rebar dimensions has proven to be a persistent challenge for researchers and practitioners. This paper introduces a semantically enriched 3D model-based system that employs computer vision and deep learning for location-aware identification and tracking of rebar issues. The system comprises four modules: (A) digital twin generation, (B) segmentation, (C) inspection, and (D) issue identification and tracking. The generation module constructs 3D models from rebar structures. The segmentation and inspection modules analyze the 3D models, enriching them with semantic information. The issue identification and tracking module exchanges information between the semantically enriched 3D models and the building information models across time. An experiment on a column rebar cage is conducted. A precision of over 90% and a recall of over 97% are reported in 3D instance segmentation. Diameter inspection achieves an accuracy of 95.5% for large-size rebars. Spacing inspection achieves a mean relative error of 0.98%. The defective spacing is identified and tracked.

## 1. Introduction

Reinforced concrete structures are the most used structures in construction as a result of their strength and economy. Furthermore, rebars play a significant role in reinforced concrete structures since they provide most of the tensile strength. To successfully execute rebar construction, it is imperative to consider key aspects, such as scheduling, quality, and cost. In terms of scheduling, enhancements of supply chain and installation planning have been undertaken [1–3]. In terms of quality, rigorous inspections must be implemented across material, processing, and installation phases. Among these inspections, rebar dimensional quality control (DQC), which aims to inspect the dimensional properties of rebars (e.g., the spacing of tie rebars and the diameter of each rebar), is probably the most critical part since bearing capacity of a structure is subjected to proper arrangement of rebars. Rebar DQC varies from project to project and is performed on construction sites, thereby posing a challenge for further investigation. With the advanced and rapid development of machine learning (ML), deep learning (DL), and computer vision (CV), these technologies offer promising solutions to traditional engineering problems [4–8]. These successful breakthroughs inspire us to develop a novel rebar dimensional quality control based on semantically enriched 3D models and building information modeling (BIM), proposing a solution to the aforementioned challenge.

The traditional approach to perform rebar DQC requires qualified inspectors and tap measures, which is time-consuming, labor-intensive, and error-prone. In recent years, several studies [7,9,10] have been conducted to improve the quality and efficiency of rebar DQC. According to them, a 3D-based system is preferred and can be divided into four major processes: (A) 3D reconstruction of rebar structures, (B) rebar geometric extraction from 3D models, (C) measurement of rebar dimensional properties, and (D) examination of measured dimensional properties. Although research on each process has been developed, certain difficulties remain when connecting the processes, such as full automation, acquisition of semantically enriched rebar 3D models, and rebar issue tracking. Below is a summary of related work:

(A) Terrestrial laser scanning (TLS) and the combination of structure from motion (SfM) and multi-view stereo (MVS) are commonly utilized to build 3D models from real assets in construction. First, since TLS is capable of scanning large structures rapidly and provides highly accurate data, it has been adopted in precast element inspection [5,7,11–14]; common structural component inspection [15–18]; rebar inspection [19–21]; mechanical, electrical, and plumbing inspection [9,22,23]; and industrial equipment inspection [24,25]. However, TLS has been found to be

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impractical on construction sites, owing to low mobility, high spatial demand, and the requirement of operators. Second, since the SfM-to-MVS pipeline shows great potential for generating 3D scenes from common 2D images, it has been applied in large-scale reconstruction from Internet photos [26–28], reinforced concrete structure inspection [6], and rebar inspection [10,29–31]. However, the relatively low accuracy of point clouds poses a challenge in dimensional inspections. Furthermore, SfM's up-to-scale point-cloud results cause a limitation in automated processes.

(B) To separate individual structural components, pipes, or rebars from a point cloud of as-built facilities, computer vision (CV) algorithms and deep learning (DL)-based approaches are possible solutions. First, random sample consensus (RANSAC) [32] and Hough transform [33] are frequently adopted in rebar inspection [20,21], pipe inspection [9,22,23], and structural component inspection [5,11,34,35]. However, RANSAC is hard to scale to large-size point clouds [16,36] and performing a Hough transform on a large dataset is computationally expensive [16]. Second, although DL-based 2D instance segmentation models such as Mask R-CNN [37] have been successfully applied in 2D rebar inspection [38,39], the segmentation results lack global spatial information. As a result, 3D segmentation models (e.g., Point Transformer [40]) may directly tackle the issue, but annotating rebar data in a 3D space is too costly. Hence, a combination of 2D segmentation and 3D fusion may be the solution. Chen et al. [31] demonstrated a prototype of a hybrid method. Nonetheless, the integration of 2D and 3D information remains an open problem.

(C) Diameter inspections of cylindrical point-cloud instances (e.g., rebars and pipes) are usually transformed into projections and 2D circle fitting. Although the geometric fitting and algebraic fitting have been applied in the literature [17,18,20,21,24,25] and show advantages in accuracy and efficiency, respectively, they both suffer from noise and outliers. Hence, the robust fitting method of the Hough transform is preferred, which has been implemented in several studies [9,22,23,34]. In spacing inspection, studies [38,41] have demonstrated automated approaches by using CV algorithms and DL. However, these studies were limited to 2D images, where information on the rebar's location is missing. Other studies [12,21,30,39] presented 3D rebar spacing inspection. Nonetheless, they were limited to rebar structures of simple arrangement.

(D) Recently, advancements in digital technology have facilitated issue identification in construction-related inspections. Studies [11,42] have utilized TLS to obtain point-cloud data of precast elements and perform dimensional measurement by CV algorithms. Then, measurements were matched and compared with the tolerances extracted from BIM models. Finally, the inspection results were stored in an Industry Foundation Classes (IFC) format. One study [9] demonstrated a similar Scan-vs-BIM method, but on pipes and conduits. Matching Scan-and-BIM is necessary to extract corresponding tolerances of as-built structures from BIM models. Maalek [15] investigated the registration of common structural components. They leveraged global registration when as-built information was provided without construction error, while global registration and subsequent local registration were integrated when construction error was contained in the data.

Despite that previous studies have shown advancements in the 3D-based rebar DQC framework, the following research gaps exist: (1) inspection algorithms are not fully automated or limited to strong assumptions (2) on-site rebar structures are hard to transform into semantic information, and (3) efficient rebar issue tracking is not explored. First, the algorithm in SfM only computes relative rotation and translation between images. The need for manual scale calibration of the reconstruction [30] hinders the 3D reconstruction from being fully automated. Also, existing studies mostly simplified the configurations of rebar installation. Specifically, they have assumed that rebars are arranged orthogonally, constrained to 2D surfaces, and positioned within well-illuminated surroundings. These stringent assumptions limit the inspection to relatively simple rebar structures, such as slabs and walls. Second, current methods include analyzing the complex rebar structures in either a 2D or 3D way. On the one hand, the 2D methods are relative mature and capable of extracting semantic attributes. However, the spatial information is lost inherently. On the other hand, the 3D methods come with a higher cost and can only be performed on well-arranged rebar structures. Third, implementing a tracking system in quality control has always been a challenging task. Despite there have been studies [8] proposing BIM-based issue tracking systems, critical issue identification and tracking still require manual handling and integration. Regarding rebar DQC, the issues to be confirmed often involve errors in the diameter of a single rebar or the spacing between two specific rebars not meeting the specifications. Linking such small-scale rebar dimensional defects with their location poses significant challenges.

Furthermore, to facilitate rebar issue tracking, digital twins (DTs), which are defined as physical products, virtual products, and connections between them [43], are incorporated in this study. DTs are expected to resolve problems regarding semantic enrichment and interoperable building representations, which are identified as the primary focuses in construction technologies [44]. Sacks et al. [45] conceptually explored digital twin construction, which can be characterized by interactions between four essential dimensions: physical–virtual, product–process, intent–status, and data–information–knowledge–decisions. Based on the inspection pipeline summarized from previous research and the concept of DTs shown in Fig. 1 (b), an ideal rebar DQC system using semantically enriched 3D model is formulated and achieves location-aware rebar issue identification. By adding the time dimension into the digital twin construction proposed by Sacks et al. [45], an efficient rebar issue tracking can be realized to connect issue identification results at different inspection checkpoints. To be clear, the connection is essentially based on consistent BIM models and registration of 3D reconstructions into BIM models.

In summary, a novel rebar DQC system is proposed to overcome the aforementioned obstacles. First, this study focuses on developing a method that fuses DL and CV techniques to construct semantic 3D models of on-site rebar structures. Second, rebar issue identification and issue tracking are fulfilled through the incorporation of BIM and semantically enriched 3D models. Third, algorithms utilized in the processes are integrated to ensure that data processing works without manual interference. Furthermore, this study adopts realistic configurations of rebar installation. Specifically, rebars include arbitrary diameters, orientations, ties, couplers, lap splicing, and are positioned in outdoor environments. This facilitates the implementation of inspections on complex rebar structures, such as columns.

This paper is organized in the following order. Section 1 introduced the background, summarized past research, and explained the objectives. Section 2 elaborates on the methodology. Section 3 details the experimental validation. Section 4 highlights and discusses the experimental results. Section 5 concludes the study and provides recommendations for future work.

## 2. Methodology

### 2.1. System workflow

Fig. 1(a) shows the system workflow, which includes four modules: (A) DT generation, (B) segmentation, (C) inspection, and (D) issue identification and tracking. First, the DT generation module constructs a 3D model from a rebar structure. Then, the segmentation module and inspection module perform analytics on the 3D model. Finally, the issue identification and tracking module exchanges information between the

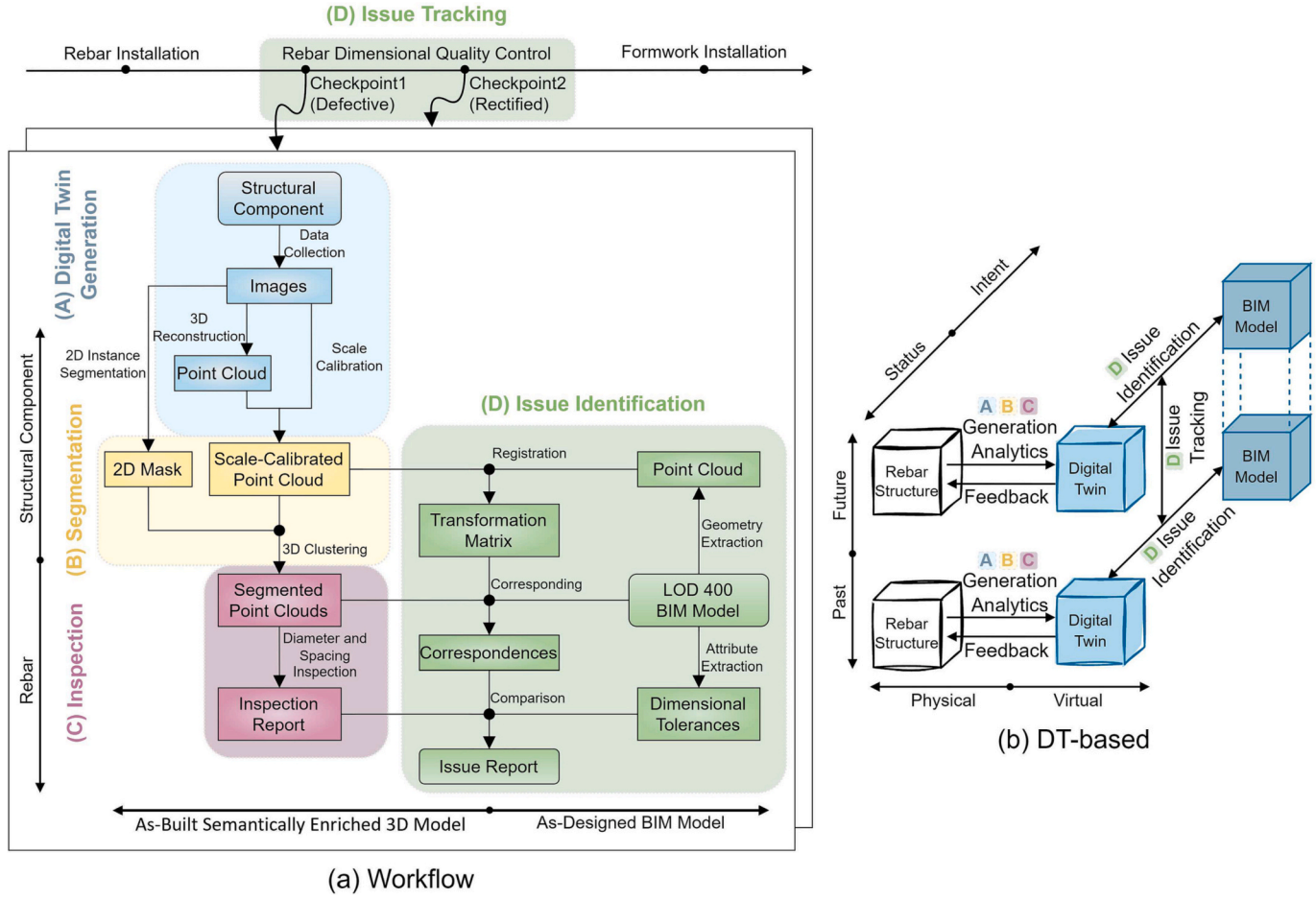



**Fig. 1.** System workflow and its interpretation as a DT-based system.

semantically enriched 3D model and the BIM model over time.

The system inputs a structural component (e.g., column and slab) and works with LOD 400 BIM models (i.e., BIM models with rebar detailing). The workflow can be characterized as (1) an autonomous process, (2) as-built vs. as-designed, and (3) a structural-component-to-rebar flow. First, the system inputs images and a BIM model, processes them with CV, DL, and BIM techniques, and outputs an issue report. The system is autonomous since no manual interference is required in the algorithms. Second, the information is extracted from the as-built semantically enriched 3D model and the as-designed BIM model. Accordingly, identifying underlying issues is accomplished by comparing the as-built and the as-designed parts. Third, a structural-component-level model is constructed and decomposed into a rebar-level model. Ultimately, the semantically enriched 3D model and the BIM model comparison is at the rebar level.

Fig. 1 (b) shows that the proposed system is fundamentally a DT-based framework that includes three conceptual dimensions: (1) physical–virtual, (2) past–future, and (3) status–intent (adapted from [45]). The DT generation and analytical processes (i.e., segmentation and inspection) are performed at different checkpoints. In this study, different checkpoints indicate whether the inspection is conducted when the rebar structure is defective or rectified. The semantic as-built 3D model is then compared with the as-designed BIM model to identify underlying issues. To incorporate results from different checkpoints, the issue-tracking process is utilized. We note that the aforementioned "as-built 3D model" is actually the "status DT" and the "as-designed BIM model" is actually the "intent DT." Hence, the proposed system is based on two types of DT.

### 2.2. Digital twin generation

Generating 3D models from 2D images is preferred since the equipment required to collect 2D data is more convenient and applicable to a construction site than that used to collect 3D data, such as TLS and Lidar. The process is divided into data collection and 3D reconstruction, which collects 2D images and reconstructs 3D point clouds from 2D images.

#### 2.2.1. Data collection

A sports camera, GoPro Hero 9 Black, is utilized to collect 2D images on a construction site, thanks to its high mobility, flexible shooting mode, and high resolution. A worker can walk on a construction site while holding the camera without restrictions. Either interval shooting or video recording is suitable for collecting the required data. The maximum 5K resolution is expected to meet the requirement of inspecting small-size objects.

During the shooting, the camera's movement, the movement speed, and the distance between the camera and the rebar structure are carefully chosen. Images are collected in spatial order to retrieve more precise camera poses when sequential matching is utilized during SfM. The camera is moved slowly at around 5 cm to 10 cm per frame to ensure a sufficient area of overlap in subsequent photos. In addition, the camera is kept close at around 30 cm to 50 cm to the rebar structure to ensure adequate pixels of rebars.

#### 2.2.2. 3D reconstruction

The generation of a 3D point cloud from 2D images is accomplished by the SfM-to-MVS pipeline and a proposed scale calibration algorithm. First, SfM is utilized to reconstruct the intrinsic and extrinsic parameters

of the camera for each image. A pixel-wise correspondence search with known camera poses generates depth maps, a point cloud, and a mesh. In this study, COLMAP [46,47] is utilized to perform the reconstruction. Exhaustive feature matching is adopted rather than sequential feature matching since it provides more accurate results at the rebar level. Although photometric and geometric consistency are utilized in COLMAP, the depth maps are complete at the structural-component level but not the rebar level. Hence, generating depth maps from a mesh eliminates the discontinuity of rebars in depth maps.

Commonly, scale ambiguity in the SfM coordinate system is overcome by selecting artificial objects with known sizes, such as predefined surveying points. Although this measure can be easily implemented, it requires manual interference. Hence, this study proposes an automated scale calibration algorithm using AprilTag [48,49], which is an efficient and robust fiducial system. Fig. 2 presents the proposed scale calibration approach. A single AprilTag of a known family is placed on the structural component at an arbitrary position. During one recording process, it is sufficient to capture a AprilTag only once. For example, when recording a slab and the surrounding columns and walls, only one AprilTag is needed. Furthermore, the AprilTag can appear only once in several frames. Then, the images containing AprilTags are collected and processed by the proposed algorithm. Finally, the scale factor is calculated to calibrate the SfM-based point cloud.

### 2.3. Segmentation

The rebar structure 3D models should be segmented into smaller rebar instances to perform a further inspection. Considering the accuracy and feasibility of acquiring 3D annotations, a hybrid model combining 2D DL and CV is proposed and illustrated in Fig. 3. The hybrid model is divided into (1) a DL-based 2D instance segmentation model to obtain 2D segmentation masks of rebar spacings and (2) a 3D clustering algorithm to integrate 2D masks of the same rebar from different viewpoints into a unique 3D instance.

#### 2.3.1. 2D instance segmentation

Mask R-CNN [37] is adopted to conduct 2D instance segmentation. Furthermore, a rebar segmentation model [38] that is trained on an on-site rebar dataset is utilized as a pre-trained model. Fine-tuning is performed, owing to the difference in the targeted domain.

Datasets of 155, 27, and 10 images are used in the training, validation, and testing. While the images include columns, slabs, and walls, this study focuses on inspecting columns. The labeling categories are formulated as spacing and intersection, which are shown in the middle of Fig. 3. This is designed to conform with the nature of rebar arrangement and to facilitate spacing inspection. First, in common structural components, vertical and horizontal rebars naturally form grids. In consideration of the occlusion due to rebar arrangement, recognizing a spacing segment as the first step is more intuitive than recognizing complete rebar. Second, a spacing instance provides information about the relationship between its ends, which is useful in spacing inspection.

#### 2.3.2. 3D clustering

A 3D clustering algorithm using CV techniques is developed to cluster 2D masks into 3D line instances automatically. The process is divided into (1) preprocessing of 2D masks, (2) clustering of raw instances, and (3) clustering of spacing instances. First, 2D masks were back-projected to 3D and are named “raw instances.” After the raw instances were down-sampled, their orientations and center lines were estimated by 3D line RANSAC and 2D circle Hough transform, respectively.

Second, Density-Based Spatial Clustering of Applications with Noise (DBSCAN) [50] was used to cluster raw instances into non-overlapping instances named “spacing instances.” The end coordinates of the center lines were utilized during the clustering. After spacing instances were extracted, outliers were removed.

Third, DBSCAN was also used to cluster spacing instances into complete linear instances named “line instances.” The segmentation method for linear objects presented in [16] was adopted. The orientation vector and the coordinate of the projected point from an arbitrary reference point were utilized during the clustering. Finally, lap splices were separated through repeated clustering on the aforementioned grouped line instances with a lowered distance threshold in DBSCAN. Fig. 4 summarizes the differences between the aforementioned raw instance, spacing instance, line instance, and rebar.

#### 2.3.3. Semantically enriched 3D model

As a result, this module enriches the complicated 3D rebar model with substantial semantic information, thereby enabling thorough and practical inspections. After the point cloud was segmented into line instances, as depicted in Fig. 3, essential rebar attributes can be efficiently acquired. Subsequently, these attributes facilitate a range of inspections. For instance, by measuring the cross section of a line instance's points, the diameter can be obtained and checked; by measuring the center lines between two line instances, the spacing can be obtained and checked; by measuring the overlapping distance along the longitudinal axis of two line instances, it is possible to determine and validate the lap splicing length and lap splicing location.

### 2.4. Inspection

To ensure structural integrity, the dimensions of rebars should be checked, where diameter and spacing are the most important items to be inspected. Thus, a diameter inspection algorithm utilizing robust CV algorithms and a spacing inspection algorithm integrating DL and CV algorithms are developed.

#### 2.4.1. Diameter inspection

A diameter inspection algorithm using CV algorithms is developed to automatically measure each line instance's diameter. The process is demonstrated in Fig. 5 and can be divided into (1) preprocessing, (2) spacing-instance-level diameter classification, and (3) line-instance-level diameter classification.

The preprocessing consists of inpainting, down-sampling, and

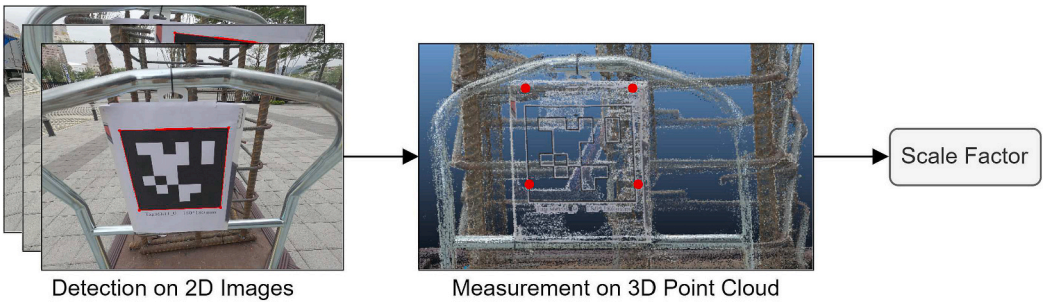



**Fig. 2.** Illustration of fiducial-marker-based scale calibration.

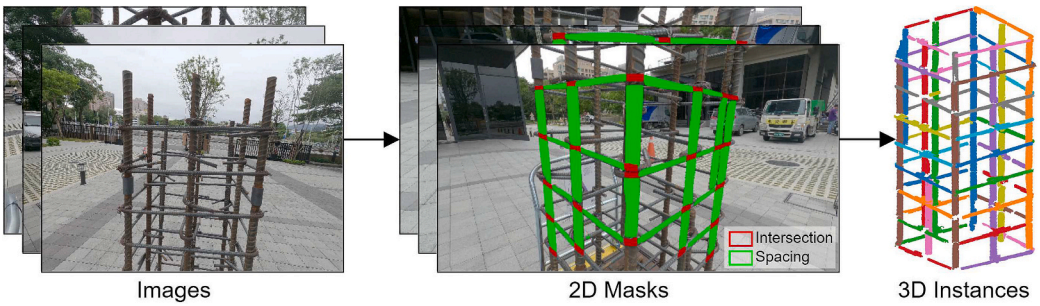



**Fig. 3.** Illustration of a hybrid model for point-cloud segmentation. At the "3D instances" step, a color map indicates different segmented instances.

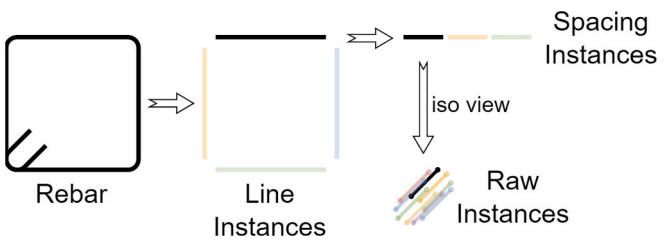



**Fig. 4.** Geometric differences between rebar, line, spacing, and raw instances. A rebar may consist of multiple line instances, a line instance may consist of multiple spacing instances, and a spacing instance may consist of multiple raw instances.

projection from 3D to 2D. Since only the outer faces of rebars are recognized by the 2D instance segmentation model, spacing instances are incomplete and should be inpainted. The rough center line of the spacing instance was first estimated. Then, points within an extended radius from the center line were resampled from the overall point cloud. After inpainting, the resampled instances were down-sampled and projected from a 3D space to a 2D plane.

The measurement of diameter is formulated as a classification problem that categorizes each line instance into the best-fitting diameter. This is accomplished by circle fitting, and a 2D circular Hough transform is utilized because of its robustness. Diameters of D10 to D32 rebars, i.e., 9.53, 12.7, 15.9, 19.1, 22.2, 25.4, 28.7, and 32.2 mm, are adopted since they are the most commonly used sizes in Taiwan's construction industry. A Hough transform was conducted for each predefined diameter, outputting the number of inliers and the estimated center. Then, the spacing instance was classified into the diameter with the maximum number of inliers. Finally, the diameter of the line instance was determined by a majority voting on the contained spacing instances.

### 2.4.2. *Spacing inspection*

A spacing inspection algorithm using CV algorithms and DL is developed to automatically detect spacings on arbitrary axes and measure the spacing distances. The process is demonstrated in Fig. 6 and can be divided into (1) clustering of line instances and (2) spacing detection and measurement. First, to fulfill the automated spacing detection, the meaning of spacing is revisited, and it can be geometrically defined as the relationship between two line instances in the same orientation. Intuitively, line instances with nearly identical orientations were clustered into groups by using DBSCAN (Fig. 6(b)).

Second, whether each pair of line instances in a group should be linked was determined by checking if any spacing instance lies in the space between the pair (Fig. 6(c)). In other words, a spacing instance that lies between a pair indicates the pair is related. To implement the strategy, the line instances in a group were projected onto 2D. Then, the following criteria were examined to determine if the relationship of a pair exists:

1. 2D overlapping ratio of the pair and the spacing instances;
2. Included angle between the pair and the spacing instances;
3. Nearest neighbors of the spacing instances; and
4. Height difference (the z-value after projection) between the pair and the spacing instances.

Finally, the spacing distance of a detected pair was measured by calculating the distance between two center lines.

### 2.5. *Issue identification and issue tracking*

To discover the dimensional defects contained in the inspection results, an issue identification algorithm utilizing BIM and CV techniques

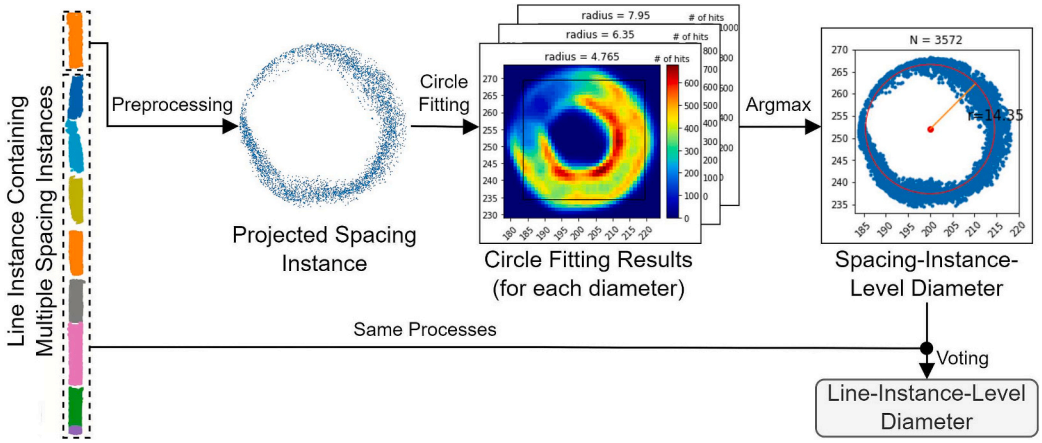



**Fig. 5.** Detailed workflow of diameter inspection. On the left side, a color map indicates different spacing instances.

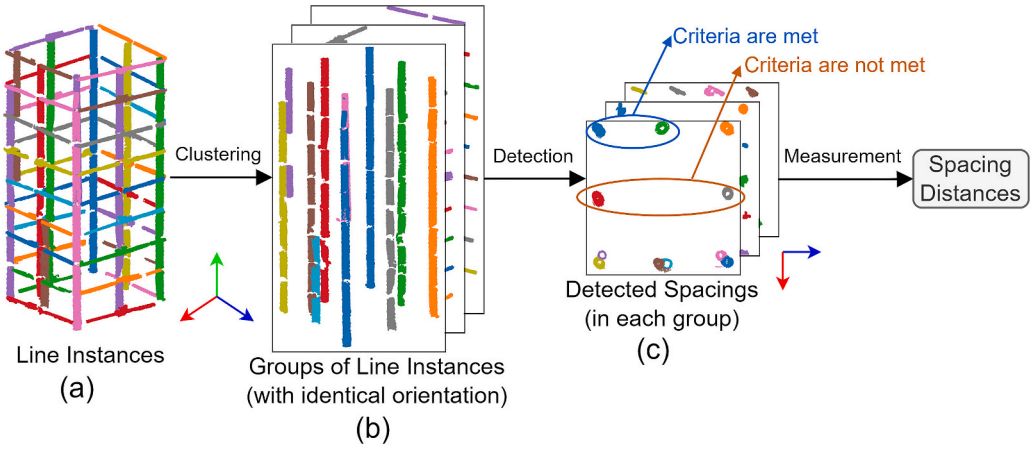



**Fig. 6.** Detailed workflow of spacing inspection. A color map indicates different line instances throughout the spacing inspection.

is proposed. In addition, to keep track of the issues until they are rectified, an issue-tracking algorithm is established. BIM is leveraged as ground truth in this module since the digitalized information facilitates automation and enables tracking over time.

### 2.5.1. Issue identification

Fig. 7 presents the concept of the proposed issue identification method. It is divided into a Scan-vs-BIM registration algorithm and a Scan-vs-BIM comparison algorithm, which constructs rebar correspondences and checks attributes in each correspondence.

The Scan-vs-BIM registration algorithm is further divided into (1) geometry extraction, (2) structural-component-level registration, and (3) rebar-level correspondence. First, Autodesk Revit was used to construct the LOD 400 BIM model and convert it to the IFC format. The geometric information of each rebar in the IFC file was interpreted and saved in a point-cloud format. Second, the 3D model and BIM point clouds were aligned by coarse and fine registrations. At the coarse (global) stage, Fast Point Feature Histogram (FPFH) [51] was utilized as a feature descriptor, and RANSAC was utilized to retrieve the best alignment in multiple iterations. At the fine (local) stage, the best transformation matrix of coarse registration was inputted as an initial constraint in the Iterative Closest Point (ICP) algorithm [52]. Iteratively, the optimum transformation was found through the cycle of finding correspondences and finding alignment. Third, after the structural components were aligned, the rebar correspondences between the 3D model and the BIM model can be recognized by querying neighboring rebars and calculating the average point distance.

The Scan-vs-BIM comparison algorithm is divided into (1) attribute extraction and (2) attribute-wise comparison. First, the rebar attributes of interest (i.e., diameter and spacing) were extracted from the LOD 400 BIM model of IFC format. A rebar attribute contains a specified value and a tolerance. Second, the inspection results in the 3D model were compared with the attributes in the BIM model. During the comparison of diameter, majority voting was performed (i.e., the diameter of a outer tie rebar is determined by the majority of diameters of four lines on each side), while during the comparison of spacing, averaging was performed. These are attributed to the fact that rebar in the BIM model may correspond to multiple lines in the 3D model.

This study utilizes LOD 400 BIM models to fulfill issue identification, which can be extended to more complicated rebar DQC such as the location of lap splices and shape of tie rebars. However, if only a basic inspection is required, such as checking rebar diameters or spacings, using LOD 300 BIM models or pure attribute information would be sufficed.

### 2.5.2. Issue tracking

An issue-tracking algorithm is proposed to record the expected rectification process of dimensional issues automatically. It is assumed that as-designed BIM models used at different checkpoints are the same, i.e., the design of the structural component remains the same. This holds in most cases since the design rarely changes at the rebar installation stage. First, the issue identification algorithm was executed on the semantic 3D models at different checkpoints. All issues were registered to the rebar instances in the BIM model. Then, an attribute-wise

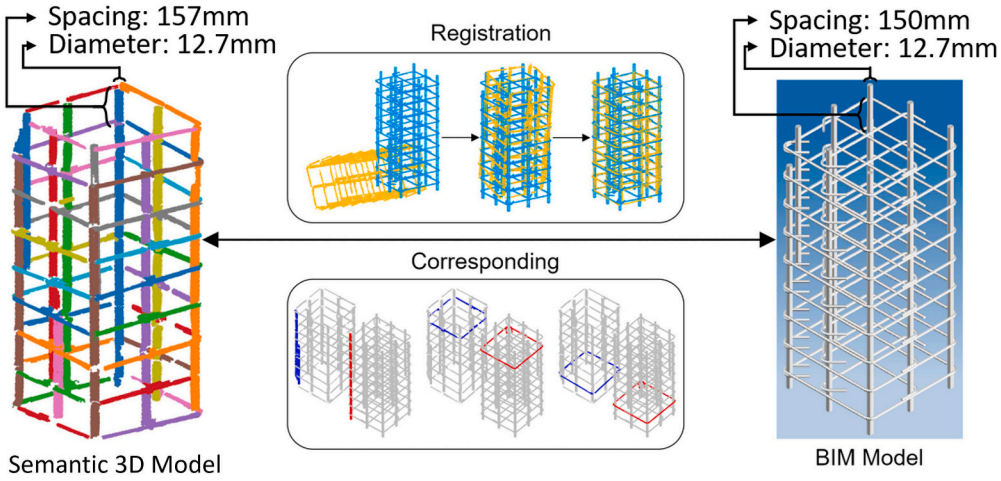



**Fig. 7.** Illustration of Scan-vs-BIM issue identification. Different colors in the semantic 3D model represent different segmented instances.

consistency check was conducted to ensure the corresponding rebars in each issue had been inspected at distinct checkpoints. Finally, the inconsistent issues were discarded, while attribute-wise issues that passed the consistency check were kept and visualized.

## 3. Experimental validation

### 3.1. Experimental setup

To validate the proposed system, an experiment on a column rebar cage was conducted to simulate the defects contained in the structure and to identify and rectify them (Section 3.2 to Section 3.5). Furthermore, the proposed system was tested on a concrete slab at a construction site. This was done to showcase its applicability when BIM models are unavailable. (Section 3.6).

Fig. 8 (a) shows the configuration of the column. It was designed to be realistic and complicated regarding diameter, shape, arrangement, and splicing. D25 and D29 were adopted for the main rebars, while D13 was adopted for the tie bars. The hook angle, hook extension length, spacing of tie bars, and size of the rebar structure all complied with common practice. Lap splicing and couplers were both adopted to increase the complexity. Fig. 8 (b) shows the two checkpoints of the experiment. At checkpoint 1, the top-layer tie rebar was moved away from the second-layer tie rebar, which forms a defective enlarged spacing, while at checkpoint 2, the rebar arrangement conformed with the project requirements.

### 3.2. Digital twin generation

In the DT generation module, the results of data collection, the SfM-to-MVS pipeline, and scale calibration are presented. First, at both checkpoints, images were collected from top to bottom at each angle surrounding the column. It took 136 s and 102 s to collect 273 images and 205 images, respectively, with the interval shooting mode at each checkpoint. Second, the camera model and the camera poses were successfully generated. Fig. 9 presents the obtained camera poses and sparse point cloud on the left side. The ratio of aligned images reaches 100% in both cases. Furthermore, the depth maps, the point cloud, and the mesh were successfully generated. Fig. 9 presents the obtained dense point cloud on the right side. Third, all AprilTags contained in the collected images were detected in both cases. Corners and orientations of the AprilTags were extracted. Subsequently, the scale factor was calculated to calibrate the point cloud. Manual measurement on the point cloud is utilized as the ground truth to evaluate the proposed algorithm. The average relative error of the scale factor is only 0.25%, which should be sufficiently accurate for construction-related activities.

### 3.3. Segmentation

In the segmentation module, the results of 2D instance segmentation and 3D clustering are presented. First, the 2D instance segmentation

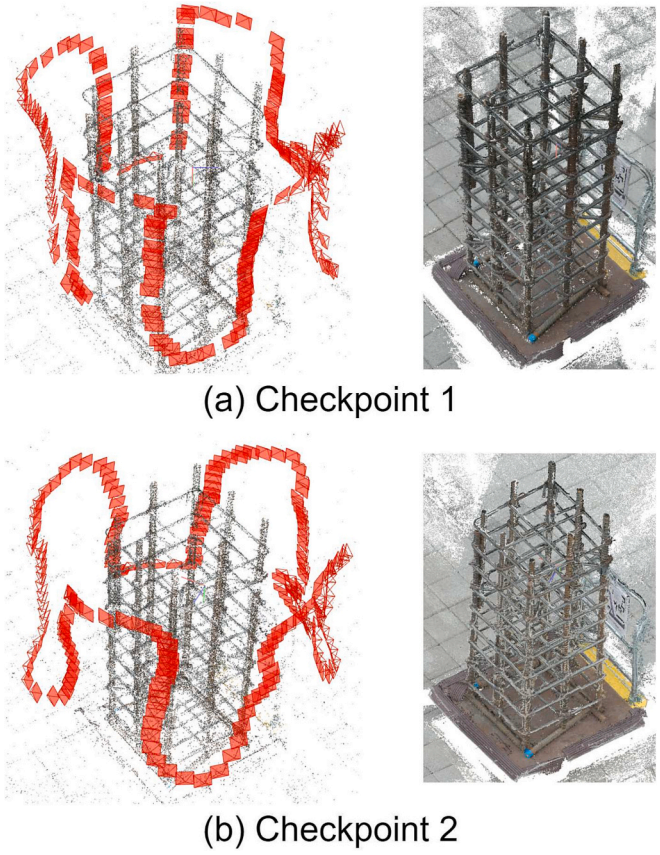



**Fig. 9.** Visualization of 3D reconstruction. (a) shows the camera poses (presented as red rectangles) and sparse point clouds (presented as gray dots) on the left and dense point clouds are presented on the right at checkpoint 1. (b) shows the camera poses (presented as red rectangles) and sparse point clouds (presented as gray dots) on the left and dense point clouds are presented on the right at checkpoint 2. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

model was established by fine-tuning the pre-trained model. It started to converge after around 5000 iterations from the pre-trained weights. Average precisions (AP) [53] of 54.1%, 50.1%, and 50.4% were reported during training, validation, and testing on column images, respectively. The precision scores are high regarding the segmentation task and show the model performs well and steadily on experimental columns. Fig. 10 presents the predicted masks of the column. Most of the spacings and intersections were correctly predicted. Predicted masks at checkpoint 1 were slightly incomplete because the model is sensitive to the scale of the image, i.e., the shooting distance.

Second, semantic 3D line instances were obtained by the proposed 3D clustering algorithm. Fig. 11 demonstrates the 3D clustering results.

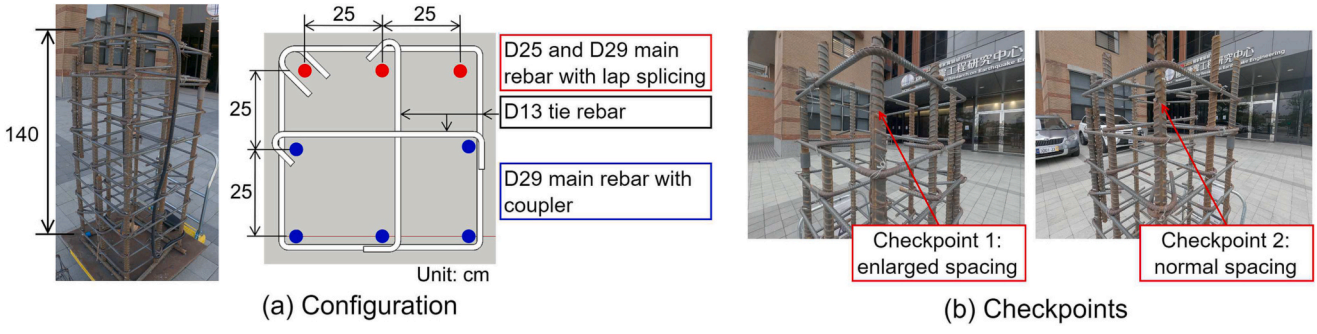



**Fig. 8.** Experimental setup of the column.

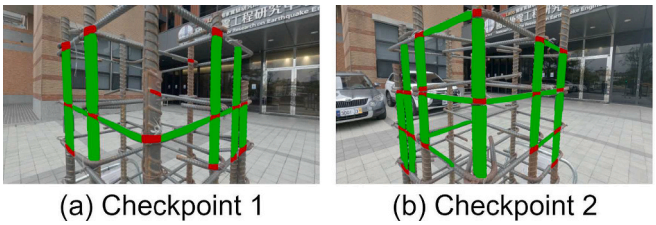



**Fig. 10.** Visualization of 2D instance segmentation. Green areas denote rebar spacing and red areas denote rebar intersections. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

In both cases, the desired line instances were successfully obtained. Taking checkpoint 2 as an example, 3323 inliers out of 3710 raw instances were clustered into 332 spacing instances. Then, 306 inliers out of 332 spacing instances were clustered into 48 line instances with lap splices separated. Table 1 presents the performance of the hybrid 3D instance segmentation model. Overall, the model achieves high precision and recall of over 90% on both main rebars and tie rebars. The precision and recall reach 100%, especially on main rebars, since larger areas in an image can mitigate the negative effects of occlusion. For instance, when processing an image from an oblique viewpoint, the predicted mask of a main rebar will be less occluded by front-layer rebars than that of a tie rebar. Accordingly, the 3D clustering would be more accurate. In addition, the lower precision compared with the recall on tie rebars is attributed to the insufficiency of the 2D model.

**Table 1**
Quantitative results of 3D instance segmentation at the line-instance level.

| Rebar Type | Precision (%) | | | Recall (%) | | |
|---|---|---|---|---|---|---|
| | Chkpt. 1 | Chkpt. 2 | Avg. | Chkpt. 1 | Chkpt. 2 | Avg. |
| Main | 100 (11/11) | 100 (11/11) | 100 | 100 (11/11) | 100 (11/11) | 100 |
| Tie | 82.9 (34/41) | 97.3 (36/37) | 90.1 | 94.4 (34/36) | 100 (36/36) | 97.2 |

Note: the number in parentheses denotes the number of instances.

### 3.4. Inspection

In the inspection module, the results of the diameter inspection and spacing inspection are presented. First, the diameter inspection results at the line and rebar levels are demonstrated in Fig. 12. Table 2 presents the quantitative results. Through a 2D circular Hough transform and Scan-vs-BIM registration, the diameters at the line and rebar levels were successfully measured. In terms of large diameter, accuracy reaches over 90% at both levels and at both checkpoints. An accuracy of over 95% is even typically reported, showing the algorithm is feasible for large-diameter rebars. In terms of small diameter, the average accuracies

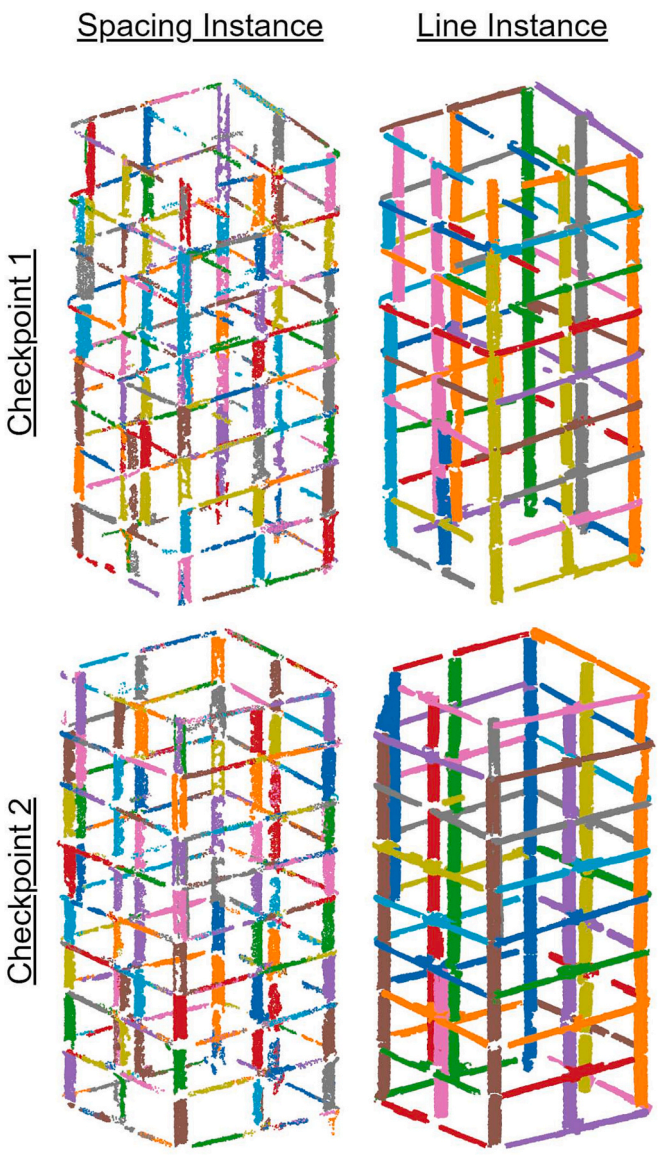



**Fig. 11.** Visualization of 3D clustering at different stages. A color map indicates different instances at each step and at each stage.

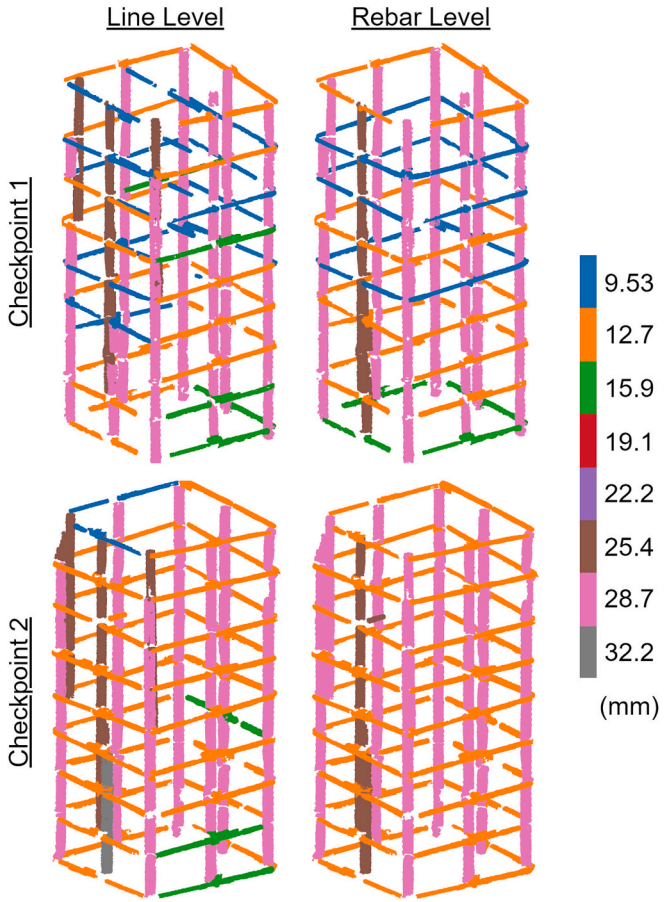



**Fig. 12.** Visualization of diameter inspection at different levels.

**Table 2**
Quantitative results of diameter inspection.

| Rebar Diameter | Accuracy (%) | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| | This study | | | | | | Kim et al. [20] | |
| | Chkpt. 1 | | Chkpt. 2 | | Avg. | | | |
| | Line Level | Rebar Level | Line Level | Rebar Level | Line Level | Rebar Level | Line Level | Rebar Level |
| D25–D32 | 100 | 100 | 90.9 | 90.9 | 95.5 | 95.5 | 65.1 | 97.2 |
| D10–D13 | 52.9 | 55.6 | 88.6 | 100 | 70.8 | 77.8 | 40.8 | 77.3 |

are 70.8% and 77.8% at the line and rebar levels, respectively. Compared with the classification accuracy of similar diameters [20], this study achieves similar performance as in previous research. However, an accurate TLS-based point cloud and an experimental slab with a simple rebar arrangement were utilized in the previous research. In contrast, a noisy SfM-based point cloud and an experimental column with a complex rebar arrangement are utilized in this study. Furthermore, in terms of small-diameter rebars, a non-negligible performance difference between the two checkpoints is observed. Further investigation shows that this is attributed to the uncertainty in 3D reconstruction. Small-size diameter inspection utilizing SfM-based point clouds remains a challenge since even noise of 1 mm in point clouds may cause significant errors in the subsequent inspections.

Second, Fig. 13 demonstrates that all groups of line instances with identical orientations were correctly detected at each checkpoint. Fig. 14 shows the measurement results of the first group at each checkpoint. Table 3 presents the quantitative performance of the spacing inspection algorithm. The mean error (ME) is 0.06 mm, showing

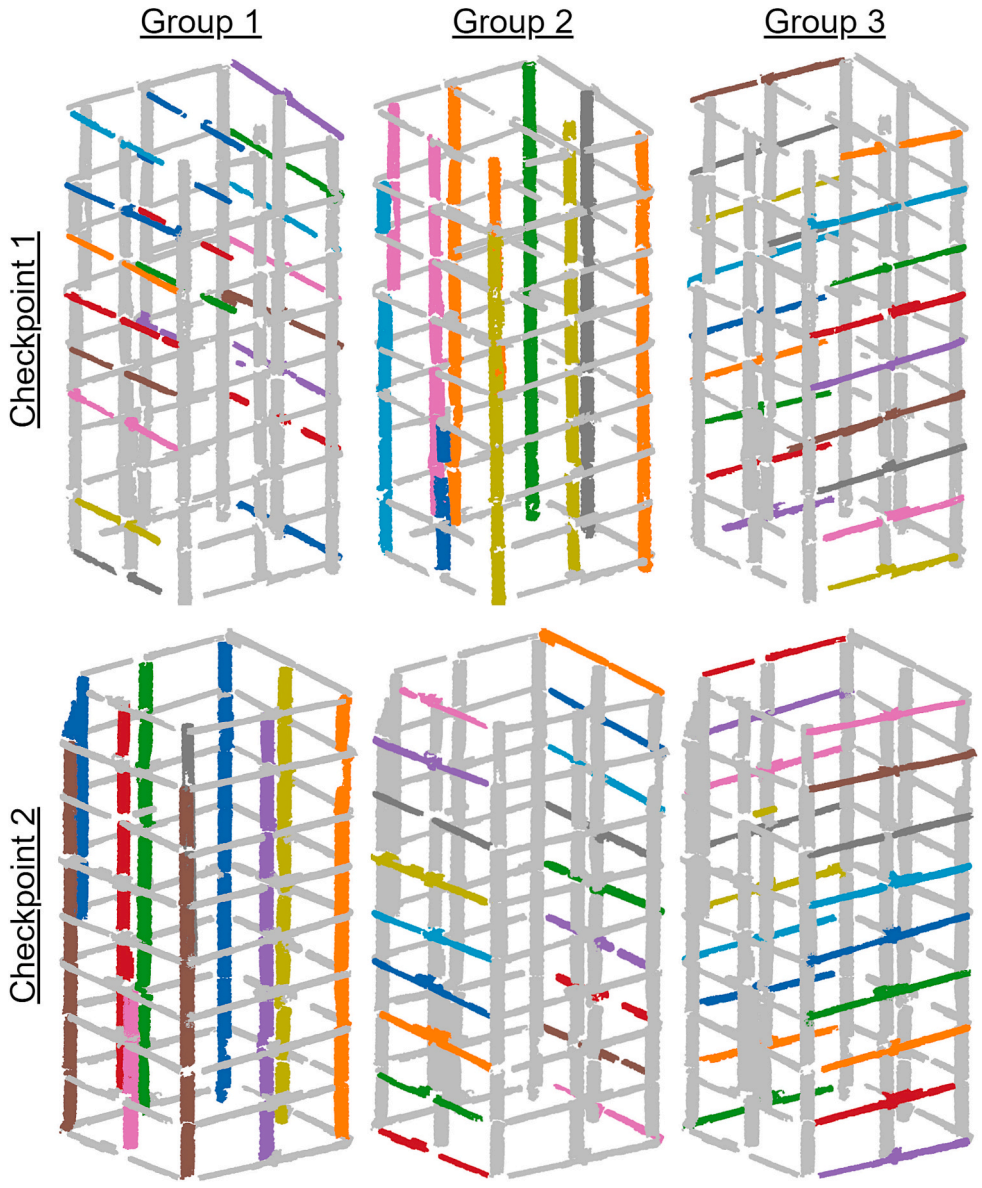



**Fig. 13.** Visualization of spacing detection. A color map indicates different instances in each group of line instances with identical orientations. Gray areas denote instances outside the group under inspection.

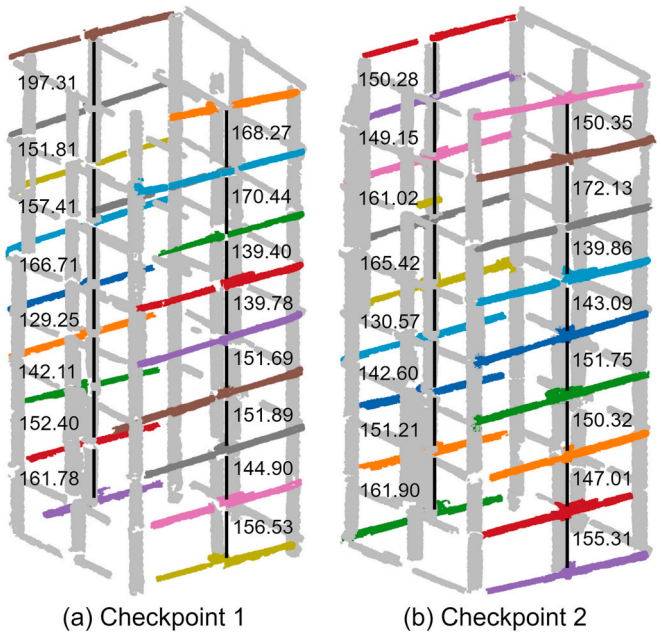



**Fig. 14.** Visualization of spacing measurement. A color map indicates different line instances in each group. Gray areas denote instances outside the group under inspection. The spacing distance is in millimeters.

**Table 3**
Quantitative results of the spacing inspection.

| | This study | | | Kim et al. [20] |
|---|---|---|---|---|
| | Chkpt. 1 | Chkpt. 2 | Avg. | |
| ME (mm) | 0.05 | 0.06 | 0.06 | – |
| MAE (mm) | 1.87 | 1.35 | 1.61 | 2.20 |
| MRE (%) | 1.11 | 0.84 | 0.98 | – |

the model is unbiased. In addition, a mean absolute error (MAE) of 1.61 mm is reported, which is lower than the 2.20 mm of a similar inspection in the literature [20]. A mean relative error (MRE) of only 0.98% indicates the proposed algorithm would be accurate for construction-related activities.

### 3.5. Issue identification and issue tracking

In the issue identification and tracking module, the results of BIM extraction, Scan-vs-BIM registration, issue identification, and issue tracking are presented. First, Table 4 demonstrates the extracted information from the LOD 400 BIM model in the IFC format. These include ID, rebar type, diameter, spacing, and spacing group ID. In addition, a geometric representation in point-cloud format was also generated for point-cloud registration.

Second, structural-component-level registration was accomplished through coarse and fine registrations. A root-mean-square error (RMSE) of 32.32 mm was achieved at the coarse stage, while a satisfying RMSE of 12.32 mm was achieved at the fine stage. However, it is shown that the registration algorithm is limited to geometrically distinct structures because it failed to align the lap splices to the correct side. During the rebar-level correspondence, the accuracy reached 100% at both checkpoint 1 and checkpoint 2 despite the existence of outlying instances and misplaced lap splices.

Finally, quality defects were identified by the proposed Scan-vs-BIM comparison algorithm (Table 5). The intentional spacing issue at checkpoint 1 was successfully identified. After dimensional quality issues at each checkpoint were identified, consistent issues across time were derived and visualized for manual interpretation (Fig. 15).

### 3.6. On-site slab experiment

In the second experiment, the rebar spacing of a slab in a RC building, which is shown in Fig. 16 (a), was conducted. The design specification of the rebar spacing for the slab is 150 mm with a tolerance of 20 mm. We sometimes only have simple design specifications in a construction site like this example. It demonstrates the applicability of the proposed system when BIM models are unavailable.

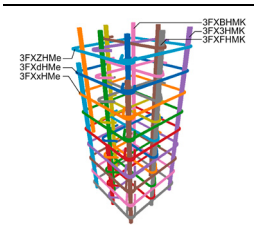
Regarding DT generation, 78 images were collected in 39 s and the ratio of aligned images reached 100%. Besides, the relative error of scale calibration was merely 1.20% compared with manual measurement. The reconstructed dense point cloud is shown in Fig. 16 (b). Regarding segmentation, 3530 raw instances extracted from the 2D segmentation masks were clustered into 57 line instances. The 3D instance segmentation model achieved a precision and recall exceeding 80%. Regarding

**Table 5**
Issue identification results. Only the first three rebars of each type are presented in Table 4.

| | Diameter (mm) | | | Spacing (mm) | | |
|---|---|---|---|---|---|---|
| ID (truncated) | BIM | Chkpt. 1 | Chkpt. 2 | BIM | Chkpt. 1 | Chkpt. 2 |
| FX3HMK | 28.7 | 28.7 | 28.7 | | | |
| FXFHMK | 28.7 | 28.7 | 28.7 | | | |
| FXBHMK | 28.7 | 28.7 | 28.7 | | | |
| FXZHMe | 12.7 | 12.7 | 12.7 | **150** | **188** | **155** |
| FXdHMe | 12.7 | 9.53 | 12.7 | 150 | 158 | 157 |
| FXxHMe | 12.7 | 9.53 | 12.7 | 150 | 152 | 152 |

Note: Bold text indicates the identified defect in spacing quality.

**Table 4**
Extracted geometric representation and attributes of the BIM model. Only the first three rebars of each type are presented for conciseness.

| Geometric Representation | Attribute | | | | |
|---|---|---|---|---|---|
| | ID (truncated) | Rebar Type | Diameter (mm) | Spacing (mm) | Spacing Group ID |
|  | 3FX3HMK | Main | 28.7 | | |
| | 3FXFHMK | Main | 28.7 | | |
| | 3FXBHMK | Main | 28.7 | | |
| | 3FXZHMe | Tie | 12.7 | 150 | 273,132 |
| | 3FXdHMe | Tie | 12.7 | 150 | 273,132 |
| | 3FXxHMe | Tie | 12.7 | 150 | 273,132 |

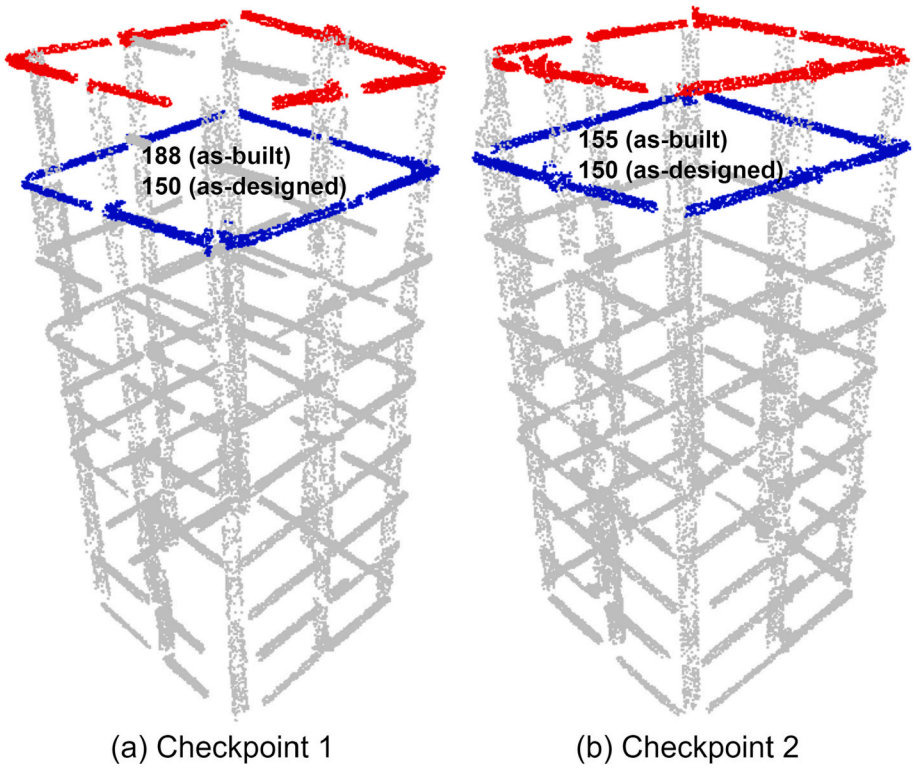



**Fig. 15.** Visualization of issue tracking of the defective rebar spacing. The spacing distance is in millimeters.

inspection, spacings on the x-axis and the y-axis were detected and measured by the proposed algorithm.

Regarding issue identification, the results of rebar axis recognition and rebar spacing comparison are presented. First, exactly two spacing groups were detected in the inspection module. By comparing spacing inspection results on each axis and conducting orthographic projection, issue identification results on both axes are visualized in Fig. 16 (c) and (d). In the Figure, green lines denote the rebars with a spacing distance that complies with design requirements, while red lines denote the rebars with a spacing distance that exceeds design tolerances (in this case, 20 mm). The values of issue identification results are tabulated in Table 6. Again, the values highlighted in red in the table exceed design tolerances. By utilizing the proposed framework, even granular issues at the mm level can be detected.

## 4. Discussion

Due to the inherent complexity, rebar DQC is conventionally conducted through manual procedures, such as measurement and interpretation of project requirements. In this study, the manual procedures are replaced with automated algorithms, such as automated scale calibration in 3D reconstruction and automated spacing detection. As a result, merely attaching fiducial markers and collecting images are manually conducted. The processing algorithms are fully automated, thereby benefiting construction management. First, video recorders could partially substitute for qualified inspectors. This reduces time, effort, and cost to accomplish rebar inspection. Second, the proposed pipeline can be further integrated with UAV, which would be the desirable inspection scheme in the long term. Third, information is digitalized, well-organized, and accessible through the DT- and BIM-based system, forming a rebar dataset and enabling later profound analytics.

In the past, semantic rebar information acquisition vastly relied on time-consuming and labor-intensive work. This study establishes a rebar 3D model that represents and synchronizes with the real rebar structure by integrating DL and CV techniques. First, a tuned SfM-to-MVS pipeline and image detection are successively utilized to build 3D models. Second, in segmentation, DL fundamentally detects rebars in complex structures, and conventional 3D clustering recovers the missing parts and removes the redundant parts. Third, the diameter inspection is achieved by combining robust 3D line detection and robust 2D circle fitting. The spacing inspection is achieved by combining 3D clustering and segmentation. The extracted semantic information enables further construction management, such as inspection and documentation.

Traditional rebar DQC depends on inspector-driven number checking between measurement results and design tolerances. This paper proposed a comparison of semantic 3D models and BIM models (i.e., status DT and intent DT). The identified issues can also be effectively tracked across time by using BIM technology. The introduced system has good potential to be extended to other domains. For instance, the DT- and BIM-based scheme may be applied to other rebar DQCs (e.g., the bending radius of hooks), other rebar inspections (e.g., binding wire tying and material inspection of rebar chairs), and other construction inspections (e.g., progress monitoring and inspection of embedded pipes).

## 5. Conclusions

This study established an autonomous system that fuses deep learning (DL), computer vision (CV), digital twins (DT), and building information modeling (BIM) to conduct novel rebar dimensional quality control. The system consists of four modules. The DT generation module collects images of a structural component using a GoPro. It reconstructs a scale-calibrated 3D point-cloud model via a structure-from-motion to multi-view stereo pipeline and an AprilTag-based algorithm. The segmentation module decomposes the 3D model via 2D instance segmentation using Mask R-CNN and 3D clustering using Density-Based Spatial Clustering of Applications with Noise. The inspection module measures the rebar diameter using a 2D circular Hough transform, while rebar spacing is measured by integrating segmentation results and CV

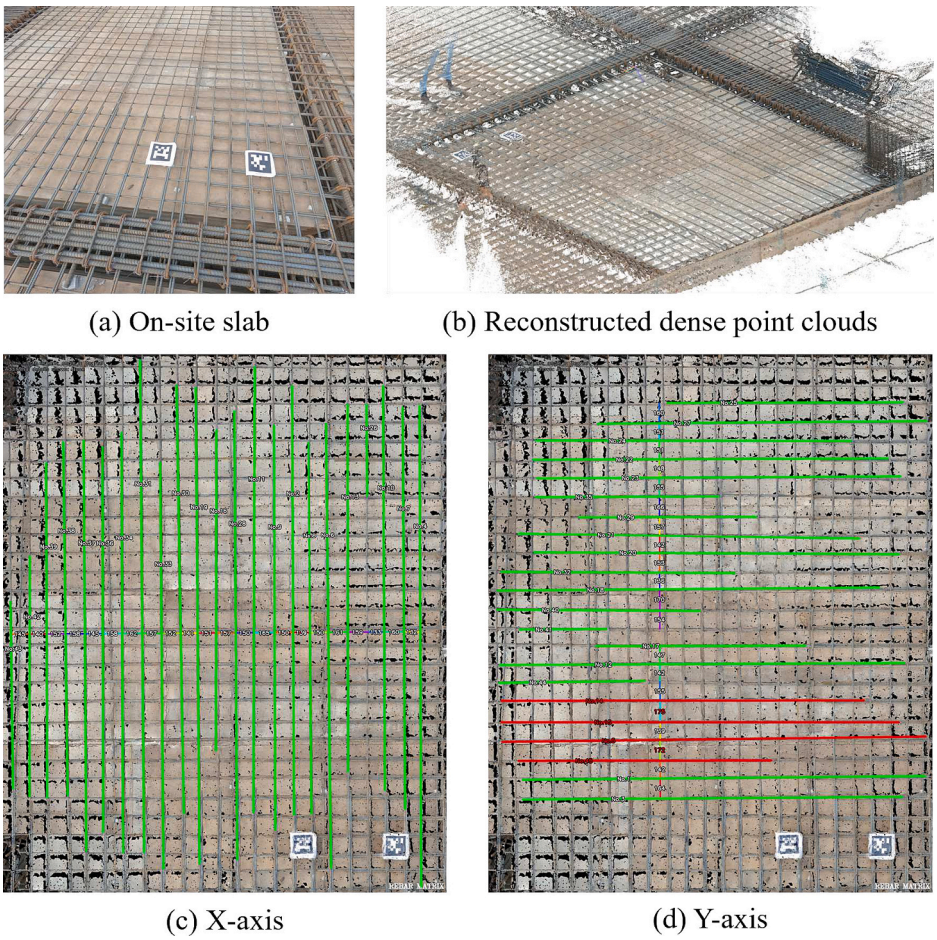



**Fig. 16.** Experimental results of the on-site slab. In (c) and (d), green denotes rebar with spacing distance that complies with project requirements; while red denotes rebar with spacing distance that exceeds tolerances. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

**Table 6**
Issue identification results of the on-site slab. The values in row X-axis correspond to the values from left to right in Fig. 16 (c). The values in row Y-axis correspond to the values from top to bottom in Fig. 16 (d).

| Spacing (mm) | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| X-axis | 145 | 140 | 152 | 158 | 145 | 158 | 162 | 157 | 152 | 141 | 151 | 157 | 150 | 165 | 150 | 139 | 150 | 161 | 159 | 133 |
| Y-axis | 160 | 151 | 151 | 148 | 155 | 166 | 157 | 142 | 153 | 136 | 170 | 154 | 147 | 142 | 155 | **175** | 139 | **172** | 142 | 164 |

Note: Bold text indicates the identified defect in spacing quality.

techniques. The issue identification and tracking module discovers the dimensional defects and keeps track of them using Scan-vs-BIM registration and comparison.

An experiment on a column rebar cage was conducted to validate the proposed system. Collected images were all aligned during 3D reconstruction. The error of scale calibration was 0.25%, which should be sufficient for construction-related tasks. The hybrid model reported an average precision of 50.4% on the column images of the test dataset in the 2D instance segmentation stage. Subsequently, it reported a precision of over 90% and a recall of over 97% on both main rebars and tie rebars in the 3D instance segmentation stage. In the case of rebar instance, the diameter inspection reached a high accuracy of 95.5% and 77.8% for large-size and small-size rebars, respectively. The spacing inspection approach achieved a low mean relative error of 0.98%. In issue identification, the proposed method reported a root-mean-square error of 12.32 mm during the structural-component-level registration and an accuracy of 100% during the rebar-level correspondence. Finally, the defective spacing was successfully identified and tracked.

In summary, three highlights of the proposed system were revealed. First, the system was built as an autonomous rebar inspection pipeline by developing automated algorithms, such as automated scale calibration. Furthermore, the autonomous pipeline benefits construction management. For example, qualified inspectors can be partially replaced with video recorders. Second, a semantic rebar 3D model was established by integrating DL and CV techniques. For instance, the combination of DL-based 2D instance segmentation and conventional 3D clustering accomplishes rebar segmentation. Consequently, accurate and robust information can be utilized for construction management, such as inspection and documentation. Third, underlying issues were effectively identified and tracked by the DT- and BIM-based information systems. This system is essentially based on Scan-vs-BIM methods, i.e., the comparison between status DT and intent DT. Furthermore, the proposed approach can be extended to other applications, such as material inspection of rebar chairs and progress monitoring.

Nevertheless, certain limitations of the proposed system have been revealed, indicating future research direction. First, unsatisfying

segmentation performance under certain conditions is observed. More diverse data, augmentation, and unsupervised training techniques can be explored to minimize incorrect predictions in 2D instance segmentation. Second, unsatisfying diameter inspection performance on small-diameter rebars is observed. DL-based approaches may be suitable to increase the accuracy since it is feasible to obtain 3D annotations for classification. DL-based point-cloud classification models, such as Point Transformer [40], are recommended to substitute for the CV-based method. Third, the diameter inspection performance may degrade when the proposed system is applied to other rebar structures (e.g., slabs and walls) due to formwork installation. Similar to the second point, DL-based approaches offer the possibility to overcome the issue.

## CRediT authorship contribution statement

**Chun-Cheng Chang:** Writing – original draft, Methodology. **Tsung-Wei Huang:** Writing – original draft, Data curation. **Yi-Hsiang Chen:** Writing – original draft, Validation, Investigation. **Jacob J. Lin:** Writing – review & editing, Supervision, Conceptualization. **Chuin-Shan Chen:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition, Conceptualization.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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